

Attacks on PRISM-id via Torsion over Small Extension Fields

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Abstract. PRISM is an isogeny-based cryptographic framework that relies on the hardness of computing a large prime-degree isogeny from a supersingular elliptic curve with an unknown endomorphism ring. It includes both an identification scheme PRISM-id and a signature scheme PRISM-sig. In this work, we present two attacks on PRISM-id. First, we analyze the probability that a randomly sampled prime q in PRISM-id results in a q -torsion subgroup defined over a small extension field, and we show that this probability is higher than claimed in the original proposal. Exploiting this observation, we construct classical forgery attacks on PRISM-id. The first attack addresses the scenario in which the attacker cannot reject a challenge. It succeeds with probability $\tilde{\Theta}(2^{-(\lambda+\log \lambda)(1-\varepsilon)})$ and runs in expected time $\tilde{O}(\max\{2^{3\lambda\varepsilon}, 2^{\lambda(\varepsilon+1/2)}\})$ for any positive real number $\varepsilon < 1/3$. Setting $\varepsilon = 1/4$ yields success probability $\tilde{\Theta}(2^{-3(\lambda+\log \lambda)/4})$ and expected time complexity $\tilde{O}(2^{3\lambda/4})$. The second forgery attack covers the scenario in which the attacker is allowed to reject challenges. It always succeeds and runs in expected time $\tilde{O}(2^{6\lambda/7})$. Finally, we describe an attack against the underlying hardness assumption of PRISM-id that runs in expected time $\tilde{O}(2^{\lambda/2})$. Note that our results do not affect the security of PRISM-sig.

Keywords: Isogenies, Post-Quantum, Identification Schemes, PRISM

1 Introduction

Isogeny-based cryptography is a promising candidate for post-quantum cryptography. A representative scheme within isogeny-based cryptography is SQIsign [1], which is one of the candidates for the NIST post-quantum additional digital signature standard. One advantage of isogeny-based cryptography is its small key size compared to other post-quantum schemes.

Many isogeny-based cryptographic protocols use supersingular elliptic curves and rely on the hardness of the *supersingular isogeny problem*, which requires computing an isogeny between two given supersingular elliptic curves. This problem is believed to be hard even in the quantum setting. Furthermore, the supersingular isogeny problem is equivalent to the problem of computing the endomorphism ring of a supersingular elliptic curve [37,31].

Recently, a new problem has been proposed and has attracted attention: computing an isogeny of a given degree d from a given supersingular elliptic curve E over $\mathbb{F}_{p^2}$. We call this the *given-degree isogeny problem*. It can be solved in polynomial time if d is smooth or if the endomorphism ring of E is known. However, when d is a large prime and the endomorphism ring of E is unknown, no efficient algorithm is known.

The given-degree isogeny problem first appeared in the context of the security of SQIsignHD [13]. SQIsignHD and several variants [4, 19, 34, 33, 38] of SQIsign including the NIST SQIsign rely on the hardness of the supersingular isogeny problem with oracle access to the given-degree isogeny problem. On the other hand, a new cryptographic framework PRISM [3] has been proposed, assuming the hardness of the given-degree isogeny problem with d being a large prime. PRISM consists of an identification scheme PRISM-id and a signature scheme PRISM-sig. Due to its simple structure, PRISM-sig could serve as a building block for more advanced cryptographic protocols.

Consequently, analyzing the hardness of the given-degree isogeny problem is important for isogeny-based cryptography. If it turns out to be easy, the security of several SQIsign variants becomes clearer because the oracle access to the given-degree isogeny problem can be removed from their security analyses. If it remains hard, the security rationale for PRISM is reinforced. However, a detailed complexity analysis of the given-degree isogeny problem is still lacking. In the case of PRISM, only rough lower-bound estimates have been provided so far.

Contributions. We analyze the hardness of the given-degree isogeny problem in the classical setting. For this, we provide algorithms required to solve the problem and give a detailed complexity analysis of each algorithm. Based on these algorithms, we construct forgery attacks on PRISM-id showing that the proposed parameters of PRISM-id do not achieve the claimed security level.

The security of PRISM-id relies on the hardness of computing an isogeny of a given large prime degree $q \approx 2^\lambda$ from a given supersingular elliptic curve E over $\mathbb{F}_{p^2}$, where λ is the security parameter. The analysis of the original proposal of PRISM claims that q -torsion points of the elliptic curve E are defined over an extension field of $\mathbb{F}_{p^2}$ of degree approximately q , and thus the attack using q -torsion points costs more than 2^λ bit operations. We analyze the probability that a randomly sampled prime q in PRISM-id results in a q -torsion subgroup defined over a relatively small extension field, and we show that an attack whose cost is less than 2^λ bit operations succeeds with probability exponentially small but greater than $2^{-\lambda}$.

Based on this observation, we construct two forgery attacks on PRISM-id. In the first attack, we assume that the attacker cannot reject a challenge prime q . This attack succeeds with probability $\tilde{O}(2^{-3\lambda/4})$ and has expected time complexity $\tilde{O}(2^{3\lambda/4})$ for a security parameter λ . In the second attack, we allow the attacker to reject a challenge prime q . This attack always succeeds and has expected time complexity $\tilde{O}(2^{6\lambda/7})$.

In addition, we analyze the hardness of the underlying problem of PRISM, which is a variant of the given-degree isogeny problem. We show that, for the

parameters of PRISM-id that target λ -bit security, there exists an algorithm with expected time complexity $\tilde{O}(2^{\lambda/2})$ that solves this problem.

Note that our attacks do not affect the security of PRISM-sig.

1.1 Organization

In Section 2, we provide background knowledge used in this paper. In Section 3, we recall the construction and security claims of the PRISM framework. In Section 4, we present the foundational algorithms and heuristics used in our attacks. In Section 5, we present two forgery attacks on PRISM-id. In Section 6, we analyze an attack on the underlying problem of PRISM-id. Finally, in Section 7, we conclude the paper.

2 Preliminaries

In this section, we provide background knowledge used in this paper.

Throughout this paper, we use the following notation. We let p be a prime number greater than 3. The symbol $\log$ denotes the natural logarithm.

2.1 Supersingular Elliptic Curves and Isogenies

An elliptic curve E defined over $\overline{\mathbb{F}}_p$ is *supersingular* if the torsion subgroup $E[p]$ is trivial. A supersingular elliptic curve is isomorphic to a curve defined over $\mathbb{F}_{p^2}$. In addition, we can choose a curve E over $\mathbb{F}_{p^2}$ such that the p^2 -th power Frobenius endomorphism on E is equal to the multiplication by $-p$. Then, for positive integers k and n such that $p \nmid n$, the group $E(\mathbb{F}_{p^{2k}})$ of rational points on E contains the n -torsion subgroup $E[n]$ if and only if $(-p)^k \equiv 1 \pmod{n}$. In particular, if n is prime, then $E[n] \subseteq E(\mathbb{F}_{p^{2(n-1)}})$ from the Fermat little theorem. We assume that all the supersingular elliptic curves considered in this paper are defined over $\mathbb{F}_{p^2}$ and have the above property. In addition, we can assume that E is a (twisted) Montgomery curve, which is defined by the equation $By^2 = x^3 + Ax^2 + x$ for some $A, B \in \mathbb{F}_{p^2}$. Thus, we assume that a supersingular elliptic curve E is represented by its coefficients A, B .

An *isogeny* between elliptic curves is a non-constant morphism of algebraic varieties that is also a group homomorphism. The *degree* of an isogeny φ is the degree of the corresponding morphism and is denoted by $\deg \varphi$. We say φ is a *d-isogeny* if $\deg \varphi = d$. Let $\varphi : E \rightarrow E'$ be a d -isogeny. Then, there exists a unique isogeny $\varphi' : E' \rightarrow E$ such that $\varphi' \circ \varphi$ is the multiplication by d on E . We call this isogeny the *dual isogeny* of φ and denote it by $\hat{\varphi}$. Assume that the degree d of φ decomposes as $d = d_1 \cdots d_k$. Then, there exist isogenies φ_i of degree d_i such that $\varphi = \varphi_k \circ \cdots \circ \varphi_1$. The order of the kernel of an isogeny φ is less than or equal to $\deg \varphi$, where the equality holds if and only if φ is separable. Conversely, for an elliptic curve E over $\overline{\mathbb{F}}_p$ and a finite subgroup G of $E(\overline{\mathbb{F}}_p)$, there exists a separable isogeny φ with kernel G , which is unique up to post-composition by an isomorphism.

An *endomorphism* of an elliptic curve E is an isogeny from E to itself or the zero morphism. The set of endomorphisms of an elliptic curve E forms a ring $\text{End}(E)$ by the addition on the images of endomorphisms and the composition of endomorphisms. We call this ring the *endomorphism ring* of E and denote it by $\text{End}(E)$.

2.2 Quaternion Algebras

In this paper, we only consider quaternion algebras over $\mathbb{Q}$ ramified exactly at p and the infinite place. If p is fixed, such an algebra is unique up to isomorphism, so we denote it by $B_{p,\infty}$. There exist $i, j \in B_{p,\infty}$ such that $(1, i, j, ij)$ is a basis of $B_{p,\infty}$ as a $\mathbb{Q}$ -vector space, i.e.,

$$B_{p,\infty} = \mathbb{Q} + \mathbb{Q}i + \mathbb{Q}j + \mathbb{Q}ij,$$

and that $i^2 = -q$, $j^2 = -p$, and $ij = -ji$, where $q = 1$ if $p \equiv 3 \pmod{4}$, $q = 2$ if $p \equiv 5 \pmod{8}$, and q is a prime such that $q \equiv 3 \pmod{4}$ and $\left(\frac{q}{p}\right) = -1$ otherwise.

The *canonical involution* on $B_{p,\infty}$ is defined by

$$\overline{a + bi + cj + dij} = a - bi - cj + dij,$$

for all $a, b, c, d \in \mathbb{Q}$. For $\alpha \in B_{p,\infty}$, the *reduced norm* of α is defined by $\text{nrd}(\alpha) = \alpha\bar{\alpha}$. A *lattice* in $B_{p,\infty}$ is a $\mathbb{Z}$ -submodule of $B_{p,\infty}$ of rank 4. An *order* in $B_{p,\infty}$ is a lattice that is a subring of $B_{p,\infty}$. Let $\mathcal{O}$ be an order in $B_{p,\infty}$. A left $\mathcal{O}$ -ideal is a lattice I in $B_{p,\infty}$ such that $\mathcal{O}I \subseteq I$. Let I be a left $\mathcal{O}$ -ideal. Then the set $\{\text{nrd}(\alpha) \mid \alpha \in I\} \subset \mathbb{Q}$ is a $\mathbb{Z}$ -module of rank 1, so this set has a unique positive generator. We define the *reduced norm* of I as the unique positive generator of this set and denote it by $\text{nrd}(I)$. A left $\mathcal{O}$ -ideal I is said to be *integral* if $I \subseteq \mathcal{O}$. In this paper, we only consider integral left $\mathcal{O}$ -ideals, and therefore we omit the word “integral” in the following sections. For $\alpha \in \mathcal{O}$, the reduced norm of α is a positive integer. Thus, the reduced norm of an integral left $\mathcal{O}$ -ideal I is also a positive integer.

An order in $B_{p,\infty}$ is called *maximal* if it is not properly contained in any other order in $B_{p,\infty}$. There exists a maximal order containing i and j . Examples of such maximal orders are given by Proposition 1 in [20]. These orders are called *special p -extremal* (see [26] for the definition). In the case $p \equiv 3 \pmod{4}$, the following order is a special p -extremal order:

$$\mathcal{O}_0 = \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}\frac{i+j}{2} + \mathbb{Z}\frac{1+ij}{2}.$$

The Deuring correspondence. If E is a supersingular elliptic curve over $\mathbb{F}_{p^2}$, then $\text{End}(E)$ is isomorphic to a maximal order in the quaternion algebra $B_{p,\infty}$. Let $\mathcal{E}_p$ be the set of the isomorphism classes of supersingular elliptic curves over

$\mathbb{F}_{p^2}$, and let $\mathcal{O}_p$ be the set of the isomorphism (as rings) classes of maximal orders in $B_{p,\infty}$. Then, we have the following map

$$\mathcal{E}_p \rightarrow \mathcal{O}_p, \quad E \mapsto \mathcal{O} \text{ such that } \mathcal{O} \cong \text{End}(E).$$

This map is surjective and the images of two distinct elliptic curves are the same if and only if the two elliptic curves are p -th power Frobenius conjugate to each other. We call this map the *Deuring correspondence*.

Let E be a supersingular elliptic curve over $\mathbb{F}_{p^2}$, and let $\mathcal{O}$ be a maximal order in $B_{p,\infty}$ isomorphic to $\text{End}(E)$. We fix an isomorphism $\iota : \mathcal{O} \rightarrow \text{End}(E)$. For a left $\mathcal{O}$ -ideal I , we define a set

$$E[I] := \bigcap_{\alpha \in I} \ker(\iota(\alpha)).$$

Then, $E[I]$ is a finite subgroup of $E(\mathbb{F}_{p^2})$. An isogeny with kernel $E[I]$ is called the *isogeny corresponding to the ideal I* .

In the case $p \equiv 3 \pmod{4}$, the elliptic curve E_0 defined by $y^2 = x^3 + x$ over $\mathbb{F}_{p^2}$ is supersingular. The endomorphism ring $\text{End}(E_0)$ is isomorphic to $\mathcal{O}_0$ defined above. An isomorphism is given by

$$((x, y) \mapsto (-x, \sqrt{-1}y)) \mapsto \mathbf{i} \text{ and } ((x, y) \mapsto (x^p, y^p)) \mapsto \mathbf{j},$$

where $\sqrt{-1}$ is a square root of -1 in $\mathbb{F}_{p^2}$.

2.3 Kani's Lemma

Since the attack on SIKE using 2-dimensional isogenies [9,30,35] appeared, there have been widespread efforts to incorporate 2-dimensional isogenies into the design of isogeny-based cryptography. The following lemma, which is called *Kani's lemma* because it comes from Kani's work [25], is a key tool in this context.

Lemma 1 (Theorem 1 in [30]). *Let d_1 and d_2 be coprime integers and let $D = d_1 + d_2$. Let E , E_1 , E_2 , and F be elliptic curves connected by the following commutative diagram of isogenies:*

$$\begin{array}{ccc} E & \xrightarrow{\varphi_1} & E_1 \\ \varphi_2 \downarrow & & \downarrow \varphi'_2 \\ E_2 & \xrightarrow{\varphi'_1} & F, \end{array}$$

where $\deg \varphi_i = \deg \varphi'_i = d_i$ for $i \in \{1, 2\}$. Then, the map

$$\Phi = \begin{pmatrix} \varphi_1 & -\widehat{\varphi'_2} \\ \varphi_2 & \widehat{\varphi'_1} \end{pmatrix} : E \times F \rightarrow E_1 \times E_2$$

is an isogeny with respect to the natural product polarizations on $E_1 \times E_2$ and $E \times F$, and has kernel $\{([d_2]P, \varphi'_2 \circ \varphi_1(P)) \mid P \in E[D]\}$, which is isomorphic to $\mathbb{Z}/D\mathbb{Z} \oplus \mathbb{Z}/D\mathbb{Z}$.

This lemma tells us how to compute all the 1-dimensional isogenies in the component of the 2-dimensional isogeny Φ from the restriction of $\varphi'_2 \circ \varphi_1$ to $E[D]$ and the degrees d_1 and d_2 . Note that Φ can be computed from its kernel (for example by using the algorithms in [11] or [29]). In particular, when $D = 2^e$ for a positive integer e , we can use the efficient formulas proposed by Dartois, Maino, Pope, and Robert [14].

Remark 1. A higher-dimensional version of Kani's lemma is known [35, Lemma 3.6]. It can be used to represent a 1-dimensional isogeny as we will explain in Section 2.4.

2.4 Representation of Isogenies

An isogeny of degree d is a rational map consisting of polynomials of degree approximately d . If the degree d is exponential in $\log p$, which is the size of the coefficients of the elliptic curve, then we need to represent the isogeny by a more compact form. To describe such a representation, we introduce the notion of an *efficient representation* of an isogeny.

Definition 1. Let d be a positive integer and $\varphi : E \rightarrow E'$ be an isogeny of degree d between elliptic curves over $\mathbb{F}_q$. An efficient representation of φ with respect to an algorithm $\mathcal{A}$ is some data $D_\varphi \in \{0, 1\}^*$ of polynomial size in $\log q$ and $\log d$ such that for all $P \in E(\mathbb{F}_{q^k})$, $\mathcal{A}$ with input (D_φ, P) returns E' and $\varphi(P)$ in polynomial time in $\log d$ and $k \log q$.

If the degree d is smooth, then we can decompose the isogeny φ into a composition of isogenies of small degree. The sequence of the representations of these small-degree isogenies as rational maps is an efficient representation of φ .

In the case where the degree d is not smooth, we can represent the isogeny φ using Kani's lemma (Lemma 1). Assume that there exists a smooth integer n such that $n^2 > d$ and $E[n]$ is rational over $\mathbb{F}_{q^k}$, where k is a positive integer in polynomial size in $\log q$. Then, a tuple $(E, P, Q, \varphi(P), \varphi(Q), d)$, where (P, Q) is a basis of $E[n]$, is an efficient representation of φ (see Theorem 1 and Remark 1 in [35]). The computation of the isogeny by this representation requires computing an 8-dimensional isogeny in general, and is not efficient in practice.

In some special cases, we can improve the efficiency of the computation of the isogeny. Assume that there exist integers d' and a with d' odd and $a > 0$ such that $d = d'(2^a - d')$ and $E[2^a]$ is rational over $\mathbb{F}_{q^k}$ for some positive integer k in polynomial size in $\log q$. Then, a tuple $(E, E', P, Q, \varphi(P), \varphi(Q), d')$, where (P, Q) is a basis of $E[2^a]$, is an efficient representation of φ . In this case, we can decompose φ into the composition of a d' -isogeny and a $(2^a - d')$ -isogeny, and apply Kani's lemma (Lemma 1) to obtain efficient representations of these isogenies. Here, we can use the efficient method of [14] to compute a 2-dimensional isogeny. This representation is used in POKÉ [5] and PRISM [3]. We also use this representation in our attacks.

2.5 Ideal to Isogeny Translation

Translating ideals into the corresponding isogenies under the Deuring correspondence is a fundamental operation in isogeny-based cryptography.

Let E be a supersingular elliptic curve over $\mathbb{F}_{p^2}$, and $\mathcal{O}$ be a maximal order in $B_{p,\infty}$ such that $\mathcal{O} \cong \text{End}(E)$. Assume that we are given E , a basis of $\mathcal{O}$, and efficient representations of the endomorphisms of E corresponding to the basis elements. In this setting, an algorithm to compute an efficient representation of the isogeny corresponding to a given left $\mathcal{O}$ -ideal I is called an *ideal-to-isogeny algorithm*.

In this paper, we restrict our attention to the case where $\text{End}(E)$ is isomorphic to a special p -extremal order. Several polynomial-time ideal-to-isogeny algorithms are known in this case (e.g., [21,20,15,16] for using 1-dimensional isogenies, and [28,4,32] for using 2-dimensional isogenies). Note that the general case can be reduced to the special case by using the above-mentioned algorithms twice.

In PRISM, the ideal-to-isogeny algorithm in [4] is used. This algorithm is efficient when the characteristic p of the base field is of the form $p = f2^e - 1$ for integers f and e such that f is odd and $p \approx 2^e$. For details of this algorithm, we refer the reader to [4, §3.2]. We denote this algorithm by `IdealToIsogeny`. More precisely, given a left $\mathcal{O}$ -ideal I , the algorithm computes an efficient representation of the isogeny corresponding to I . We omit the elliptic curve and the special p -extremal order from the input of `IdealToIsogeny` since we regard these as system parameters.

2.6 Computational Problems

Isogeny-based protocols using supersingular elliptic curves are based on the hardness of the following computational problem.

Problem 1 (Supersingular isogeny problem). Given two supersingular elliptic curves E and E' over $\mathbb{F}_{p^2}$, find an isogeny $\varphi : E \rightarrow E'$.

One of the best algorithms for solving the supersingular isogeny problem is the Delfs-Galbraith algorithm [17], which has expected time complexity $\tilde{O}(p^{1/2})$ in classical computation and $\tilde{O}(p^{1/4})$ in quantum computation with Grover's algorithm [22].

The following related problem is also important in isogeny-based cryptography.

Problem 2 (Maximal order problem). Given a supersingular elliptic curve E over $\mathbb{F}_{p^2}$, find a maximal order $\mathcal{O}$ in $B_{p,\infty}$ such that $\mathcal{O} \cong \text{End}(E)$.

Here, finding a maximal order in $B_{p,\infty}$ means finding a basis of $\mathcal{O}$ as a tuple of $\mathbb{Q}$ -vectors with respect to the basis $(1, i, j, ij)$.

It is known that Problem 1 and Problem 2 are equivalent, i.e., there exist polynomial-time reductions between them. It was shown by assuming the generalized Riemann hypothesis in [37], and later unconditionally in [31].

Another problem relevant to isogenies between supersingular elliptic curves is computing an isogeny from a given elliptic curve E and degree d .

Problem 3 (Computing isogeny given degree problem). Given a supersingular elliptic curve E over $\mathbb{F}_{p^2}$ and a positive integer d , find an isogeny φ from E of degree d .

If d is smooth, then this problem can be solved efficiently by applying Vélú's formula [36] to each prime factor of d . Even if d is not smooth, if we are given a maximal order $\mathcal{O}$ in $B_{p,\infty}$ such that $\mathcal{O} \cong \text{End}(E)$, then we can still compute an isogeny of degree d from E by using an ideal-to-isogeny algorithm. This is a core component of PRISM and will be explained in Section 3.1. On the other hand, if d is not smooth and we are not given a maximal order $\mathcal{O}$, then there is no known efficient algorithm to solve this problem. We will give a more detailed analysis of this problem when d is a prime (see Lemma 3).

3 PRISM

In this section, we recall PRISM, which consists of an identification scheme PRISM-id and a signature scheme PRISM-sig derived from PRISM-id. The security of PRISM is based on the hardness of a variant of Problem 3.

3.1 PRISM-id

We give an overview of the identification scheme PRISM-id. We refer to the original paper [3] for further details.

First, we define the notation used in the scheme. Let p be a prime of the form $p = f2^e - 1$ for some integers f and e . We require $p \approx 2^e$ so that we can use `IdealToIsogeny` efficiently. Let a be an integer less than e , and let Primes_a be the set of all primes in $[2^{a-1}, 2^a]$.

PRISM-id is a two-round identification scheme, which is based on the following diagram.

$$\begin{array}{ccc} E_0 & & \\ \downarrow \varphi_{\text{sk}} & & \\ E_{\text{vk}} & \xrightarrow{\varphi_{\text{rsp}}} & E_{\text{rsp}} \end{array}$$

Here, E_0 is a supersingular elliptic curve whose endomorphism ring is isomorphic to a special p -extremal order $\mathcal{O}_0$.

The system parameter of PRISM-id is the tuple $(e, f, a, E_0, \mathcal{O}_0)$. The secret key for PRISM-id is a left $\mathcal{O}_0$ -ideal I_{sk} of norm ℓ^n for some prime $\ell > 2^a$ and integer n . The public key is the codomain E_{vk} of the isogeny φ_{sk} corresponding to I_{sk} . The challenge for the identification scheme is a prime q sampled uniformly at random from Primes_a . The response to the challenge is an isogeny φ_{rsp} from E_{vk} of degree $q(2^a - q)$. The verifier checks whether the response φ_{rsp} is an isogeny from E_{vk} of degree $q(2^a - q)$. The procedure of the identification scheme

Table 1. PRISM-id identification scheme.

Prover($\text{sk} = I_{\text{sk}}$)	Verifier($\text{vk} = E_{\text{vk}}$)
	Sample a prime $q \xleftarrow{\$} \text{Primes}_a$
$\varphi_{\text{rsp}} \leftarrow \text{GenIsogeny}(E_{\text{vk}}, I_{\text{sk}}, q)$	
	Return $\text{VerIsogeny}(E_{\text{vk}}, \varphi_{\text{rsp}}, q)$

is depicted in Table 1, where the algorithms **GenIsogeny** and **VerIsogeny** are defined as follows.

- **GenIsogeny**(E, I, q) $\rightarrow \varphi$: On input a supersingular elliptic curve E over $\mathbb{F}_{p^2}$, a left $\mathcal{O}_0$ -ideal I such that the codomain of the isogeny corresponding to I is E , and a prime $q \in \text{Primes}_a$, outputs a $q(2^a - q)$ -isogeny φ from E .
- **VerIsogeny**(E, φ, q) $\rightarrow \text{accept/reject}$: On input a supersingular elliptic curve E over $\mathbb{F}_{p^2}$, an isogeny φ , and a prime $q \in \text{Primes}_a$, outputs **accept** if the domain of φ is E and $\deg \varphi = q(2^a - q)$, and **reject** otherwise.

The algorithm **VerIsogeny** uses the decomposition of the response isogeny φ_{rsp} as $\varphi_{\text{rsp}} = \varphi_{2^a - q} \circ \varphi_q = \psi_q \circ \psi_{2^a - q}$, where $\deg \varphi_q = \deg \psi_q = q$ and $\deg \varphi_{2^a - q} = \deg \psi_{2^a - q} = 2^a - q$. Thus, we have the following commutative diagram.

$$\begin{array}{ccc}
 E_{\text{vk}} & \xrightarrow{\varphi_q} & E_1 \\
 \psi_{2^a - q} \downarrow & \searrow \varphi_{\text{rsp}} & \downarrow \varphi_{2^a - q} \\
 E_2 & \xrightarrow{\psi_q} & E_{\text{rsp}}
 \end{array}$$

By applying Kani's lemma (Lemma 1), we can obtain efficient representations of the isogenies in the diagram. The degrees of these isogenies can be checked by computing the 2^a -Weil pairings of the images of 2^a -torsion points under the isogenies.

3.2 PRISM-sig

PRISM-sig is a signature scheme based on PRISM-id. For constructing PRISM-sig, we prepare a collision-resistant hash function H_{prime} whose output is a prime number in Primes_a (see [3, §3.3] for construction details).

The key generation of PRISM-sig is the same as that of PRISM-id except that we take $a \approx 2\lambda$. Let E_{vk} be the public key and I_{sk} be the secret key. For a message $m \in \{0, 1\}^*$, we let $q = H_{\text{prime}}(E_{\text{vk}} || m)$. Then the signature is $\text{GenIsogeny}(E_{\text{vk}}, I_{\text{sk}}, q)$, and the verification is done by $\text{VerIsogeny}(E_{\text{vk}}, \varphi_{\text{rsp}}, q)$.

3.3 Security

In this subsection, we briefly recall the security analysis of PRISM from [3].

First, to define the underlying problem of PRISM-id, we recall the definition of the oracle SPEDIO defined in [3].

Definition 2 (Definition 5 in [3]). A special degree isogeny oracle (SPEDIO) is an oracle which takes as input a supersingular elliptic curve E and a prime $q \in \text{Primes}_a$, and returns a uniformly random isogeny of degree $q(2^a - q)$ from E .

Using this oracle, we define the underlying problem of PRISM-id.

Problem 4 (Problem 2 in [3]). Given a supersingular elliptic curve E and SPEDIO, compute an isogeny of degree $q(2^a - q)$ from E with a prime $q \in \text{Primes}_a$ different from all degrees previously queried to the oracle.

The security of PRISM-id and PRISM-sig is established based on this problem in [3] as follows.

Proposition 1 (Proposition 1 in [3]). Under the assumption that Problem 4 is hard, the security against active attacks of PRISM-id performing N interactions has a winning probability bounded by $N/\#\text{Primes}_a$.

Proposition 2 (Proposition 2 in [3]). If the hash function H_{prime} in Section 3.2 is a collision-resistant cryptographic hash function and Problem 4 is hard, then PRISM-sig is EUF-CMA secure.

Next, we recall how to choose the parameters p and a to ensure a λ -bit security for classical attacks and a $\lambda/2$ -bit security for quantum attacks, which is discussed in [3, §4.4].

One of the best methods for a key recovery attack on PRISM-id and PRISM-sig is to find an isogeny from E_0 to E_{vk} of smooth degree. We can use the Delfs-Galbraith algorithm [18] for this purpose. As we explained in Section 2.6, the expected time complexity of this algorithm is $\tilde{O}(\sqrt{p})$ in the classical setting, and $\tilde{O}(p^{1/4})$ in the quantum setting. Therefore, we require that p is at least approximately $2^{2\lambda}$.

The authors of [3] estimate the cost to compute a $q(2^a - q)$ -isogeny from E_{vk} for $q \in \text{Primes}_a$. They claim that its expected time complexity is at least $\tilde{O}(q^2)$ in the classical setting. Therefore, we require that a is at least approximately $\lambda/2$.

For PRISM-id, we require that the probability in Problem 4 is less than $2^{-\lambda}$. This leads to the requirement that $\#\text{Primes}_a > 2^\lambda$. From the prime number theorem, we have

$$\#\text{Primes}_a \approx \frac{2^a}{a \log 2} - \frac{2^{a-1}}{(a-1) \log 2} \approx \frac{2^a}{2a \log 2}.$$

Therefore, we require that $a > \lambda + \log_2 \lambda$.

For PRISM-sig, we require that the cost to find a collision in the hash function H_{prime} is greater than 2^λ . This can be achieved by ensuring that $\#\text{Primes}_a > 2^{2\lambda}$, thus $a > 2\lambda + \log_2 \lambda$. However, they claim that $a > 2\lambda$ is enough to ensure security because one computation of H_{prime} requires another cryptographic hash function several times. For more details, see [3, §4.4].

4 Foundations of Our Attacks

In this section, we give algorithms and heuristics used in our attacks. From now on, the notation $\tilde{O}$ hides logarithmic factors in p as well as in its input. Our algorithms are classical algorithms and some of these are probabilistic. We consider the expected time complexity of these algorithms, not the worst-case time complexity.

4.1 Algorithms

We introduce the following algorithms used in our attacks.

- **ConstructField**(p, k) $\rightarrow \mathbb{F}_{p^{2k}}$: On input a prime p and a positive integer k , outputs the extension field $\mathbb{F}_{p^{2k}}$, whose representation will be explained later.
- **Isogeny**($E, q, \mathbb{F}_{p^{2k}}$) $\rightarrow \varphi$: On input a supersingular elliptic curve E , a prime q , and the extension field $\mathbb{F}_{p^{2k}}$ such that $E[q] \subset E(\mathbb{F}_{p^{2k}})$, outputs an isogeny φ of degree q from E .
- **IsGoodPrime**(p, q, B) $\rightarrow \text{True/False}$: On input distinct primes p and q and a bound B , outputs a boolean indicating whether there exists a positive integer $k \leq B$ such that $(-p)^k \equiv 1 \pmod{q}$.

Constructing extension fields (ConstructField(p, k)). Since our algorithms require operations in an extension field $\mathbb{F}_{p^{2k}}$ for some exponentially large k , we need an algorithm to construct a computable representation of $\mathbb{F}_{p^{2k}}$. We also need to analyze the cost of the construction and of the operations in $\mathbb{F}_{p^{2k}}$.

Let k be a positive integer, and consider the field extension $\mathbb{F}_{p^{2k}}/\mathbb{F}_{p^2}$. We assume that $\mathbb{F}_{p^{2k}}$ is expressed as $\mathbb{F}_{p^{2k}} = \mathbb{F}_{p^2}[X]/(F(X))$, where $F(X)$ is an irreducible polynomial of degree k over $\mathbb{F}_{p^2}$. Then, the cost of addition and multiplication in $\mathbb{F}_{p^{2k}}$ is in $\tilde{O}(k)$ times the cost of addition and multiplication in $\mathbb{F}_{p^2}$. This cost is achieved by using the polynomial multiplication algorithm [12, Algorithm 9.6.1] and polynomial modulo algorithm [12, Algorithm 9.6.4] with FFT integer multiplication (see [12, §9.5.8]).

Suppose that k is the product of $d_1, \dots, d_n$, where d_i are positive integers. Consider the tower of field extensions

$$\mathcal{F}_0 = \mathbb{F}_{p^2} \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots \subset \mathcal{F}_n = \mathbb{F}_{p^{2k}},$$

where the extension degree of $\mathcal{F}_i$ over $\mathcal{F}_{i-1}$ is d_i . Similarly to the above, we assume that $\mathcal{F}_i = \mathcal{F}_{i-1}[X]/(F_i(X))$, where $F_i(X)$ is an irreducible polynomial

of degree d_i over $\mathcal{F}_{i-1}$. Then, by the same discussion, we can show that the cost of addition and multiplication in $\mathbb{F}_{p^{2k}}$ is in $\tilde{O}(k)$.

The cost of inversion in $\mathbb{F}_{p^{2k}}$ is in $\tilde{O}(k^2)$ since the inversion can be done by the power of $p^{2k} - 2$ in $\mathbb{F}_{p^{2k}}$. The expected cost of square root in $\mathbb{F}_{p^{2k}}$ is in $\tilde{O}(k^2)$ by using a variant of the Tonelli-Shanks algorithm [2, Algorithm 5].

The remaining problem is how to construct $\mathbb{F}_{p^{2k}}$. To deal with this problem, we use the following lemma.

Lemma 2. *Let ℓ be a prime number, and i be a positive integer. Suppose that we are given a representation of $\mathbb{F}_{p^{2i}}$ by a tower of field extensions with the base field $\mathbb{F}_{p^2}$. Then there exists an algorithm with input ℓ and $\mathbb{F}_{p^{2i}}$ returning an irreducible polynomial of degree ℓ over $\mathbb{F}_{p^{2i}}$ having expected time complexity $\tilde{O}(\ell^3 \cdot i^2)$.*

Proof. We consider the following algorithm:

1. Choose a random monic polynomial $F(X) \in \mathbb{F}_{p^{2i}}[X]$ of degree ℓ .
2. Check whether $F(X)$ is irreducible over $\mathbb{F}_{p^{2i}}$.
3. If $F(X)$ is irreducible, return $F(X)$.
4. Otherwise, go to step 1.

The expected time complexity of step 1 is in $\tilde{O}(\ell \cdot i)$ since it can be done by generating $\ell \cdot i$ random elements in $\mathbb{F}_{p^2}$.

From Proposition 3.4.4 in [10], $F(X)$ is irreducible over $\mathbb{F}_{p^{2i}}$ if and only if $X^{p^{2\ell \cdot i}} \equiv X \pmod{F(X)}$ and $\gcd(X^{p^{2i}} - X, F(X)) = 1$. Let $r_1(X), r_\ell(X)$ be the representatives of the classes $X^{p^{2i}} \pmod{F(X)}$ and $X^{p^{2\ell \cdot i}} \pmod{F(X)}$ of degree less than ℓ , respectively. The expected time complexity of computing $r_\ell(X)$ is in $\tilde{O}(\ell^2 \cdot i^2)$. Thus, the expected time complexity of checking $X^{p^{2\ell \cdot i}} \equiv X \pmod{F(X)}$ is in $\tilde{O}(\ell^2 \cdot i^2)$. The expected time complexity of computing $r_1(X)$ is in $\tilde{O}(\ell \cdot i^2)$. The expected time complexity of computing $\gcd(r_1(X), F(X))$ is in $\tilde{O}(\ell^2 \cdot i)$. Therefore, the expected time complexity of step 2 is in $\tilde{O}(\ell^2 \cdot i^2)$.

A well-known result by Gauss (see [27, Exercise V.22]) tells us that the number of irreducible polynomials of degree ℓ over $\mathbb{F}_{p^{2i}}$ is $(p^{2\ell \cdot i} - p^{2i})/\ell$. Therefore, the probability that a random monic polynomial of degree ℓ over $\mathbb{F}_{p^{2i}}$ is irreducible is approximately $1/\ell$. Thus, the expected number of iterations of the above algorithm is approximately ℓ . Therefore, the expected time complexity of the algorithm is in $\tilde{O}(\ell^3 \cdot i^2)$. $\square$

Now, we deal with the construction of $\mathbb{F}_{p^{2k}}$. Decompose k into prime factors $k = \ell_1 \cdots \ell_n$, where ℓ_i are (not necessarily distinct) prime numbers. We assume that ℓ_n is the largest prime factor of k . For $i = 1, \dots, n$, we define $q_i = p^{2\ell_1 \cdots \ell_i}$ and $q_0 = p^2$. Then, we construct $\mathbb{F}_{p^{2k}}$ by the tower of field extensions

$$\mathbb{F}_{p^2} = \mathbb{F}_{q_0} \subset \mathbb{F}_{q_1} \subset \cdots \subset \mathbb{F}_{q_n} = \mathbb{F}_{p^{2k}}.$$

For each $i = 1, \dots, n$, we construct $\mathbb{F}_{q_i}$ from $\mathbb{F}_{q_{i-1}}$ by finding an irreducible polynomial $f_i(X)$ of degree ℓ_i over $\mathbb{F}_{q_{i-1}}$. From Lemma 2, the expected time

complexity of this procedure is $\tilde{O}(\ell_i^3(\ell_1 \cdots \ell_{i-1})^2)$. Therefore, the total expected time complexity is

$$\tilde{O}\left(\sum_{i=1}^n \ell_i^3(\ell_1 \cdots \ell_{i-1})^2\right) \leq \tilde{O}\left(n \cdot \max_{i=1, \dots, n} \{\ell_i^3(\ell_1 \cdots \ell_{i-1})^2\}\right)$$

Since $n \leq \log_2 k$ and $\ell_i^3(\ell_1 \cdots \ell_{i-1})^2 \leq k^3$, the total expected time complexity is $\tilde{O}(k^3)$. We denote the algorithm to construct a computable representation of $\mathbb{F}_{p^{2k}}$ by `ConstructField`(p, k).

Computing q -isogenies (`Isogeny`($E, q, \mathbb{F}_{p^{2k}}$)). Next, we give an algorithm to compute a q -isogeny from a supersingular elliptic curve E for a prime q different from p .

Lemma 3. *Let E be a supersingular elliptic curve, q be a prime different from p , and k be a positive integer such that $(-p)^k \equiv 1 \pmod{q}$. Then there exists an algorithm with input E, q , and a computable representation of $\mathbb{F}_{p^{2k}}$ to compute (an efficient representation of) a q -isogeny with expected time complexity $\tilde{O}(\max\{k^2, k\sqrt{q}\})$.*

Proof (Special case). For simplicity, we prove the case where p and q satisfy the condition in PRISM. This case is sufficient for our attacks. In particular, we assume that $p = f2^e - 1$ for some odd integer f and $e \geq 2$, and that $q \in \text{Primes}_a$ for some $a < e$. We give a proof for the general case in Appendix A.

We can assume that E is a Montgomery curve since a transformation to a Montgomery curve can be done in polynomial time. We consider the following algorithm to compute a q -isogeny φ from E .

1. Take a random point $P \in E(\mathbb{F}_{p^{2k}})$.
2. Compute the scalar multiplication $[((-p)^k - 1)/q]P$. Let Q be the result.
3. If $Q = 0$ then return to step 1.
4. Let φ be the isogeny of degree q with kernel $\langle Q \rangle$ and (P_a, Q_a) be a basis of $E[2^a]$. Compute the codomain E' and images $\varphi(P_a)$ and $\varphi(Q_a)$ by using the $\sqrt{\text{élu}}$'s formula by [6].
5. Output $(E', q, P_a, Q_a, \varphi(P_a), \varphi(Q_a))$ as an efficient representation of φ .

The point Q in step 2 has order q or 1 since $E(\mathbb{F}_{p^{2k}}) = E[(-p)^k - 1]$ and q is a prime. The output is an efficient representation of φ since $q < 2^a$. Therefore, the algorithm is correct.

In the following, we analyze the time complexity of each step. Note that these steps are done in the x -coordinates-only arithmetic on the Montgomery curve.

Step 1 consists of sampling a random element in $\mathbb{F}_{p^{2k}}$ as the x -coordinate of P and checking whether the corresponding y -coordinate is in $\mathbb{F}_{p^{2k}}$. The former requires k times random samplings in $\mathbb{F}_{p^2}$. The latter requires a constant number of additions and multiplications in $\mathbb{F}_{p^{2k}}$ and a square detection in $\mathbb{F}_{p^{2k}}$, which can be done by powering to $(p^{2k} - 1)/2$. These procedures are repeated until a

$\mathbb{F}_{p^{2k}}$ -rational point P is found. The expected number of iterations is 2. Therefore, the expected time complexity of step 1 is in $\tilde{O}(k^2)$.

Step 2 consists of $O(k)$ times differential additions and doublings on $E(\mathbb{F}_{p^{2k}})$. The differential addition and multiplication require a constant number of additions and multiplications in $\mathbb{F}_{p^{2k}}$. Therefore, the expected time complexity of step 2 is in $\tilde{O}(k^2)$.

The failure probability of step 3 is $1/q$. Therefore, we can expect that several iterations of steps 1–3 are sufficient to obtain a non-trivial point Q .

The $\sqrt{\text{elu}}$'s formula in step 4 requires $\tilde{O}(\sqrt{q})$ additions and multiplications in $\mathbb{F}_{p^{2k}}$. Therefore, the expected time complexity of step 4 is in $\tilde{O}(k\sqrt{q})$. This completes the proof. $\square$

We denote the algorithm in Lemma 3 by $\text{Isogeny}(E, q, \mathbb{F}_{p^{2k}})$.

Checking the rationality of the q -torsion subgroup ($\text{IsGoodPrime}(p, q, B)$).

Finally, we give an algorithm to determine whether the q -torsion subgroup of a supersingular elliptic curve is rational over some extension field $\mathbb{F}_{p^{2k}}$ for $k \leq B$ for a given prime q and a positive integer B . As we explained in Section 2.1, the q -torsion subgroup of a supersingular elliptic curve is rational over $\mathbb{F}_{p^{2k}}$ if and only if $(-p)^k \equiv 1 \pmod{q}$ in our setting.

Lemma 4. *Let q be a prime different from p and B be a positive integer. Then there exists an algorithm with input q and B to determine whether q divides $\prod_{k=1}^B ((-p)^k - 1)$ whose expected time complexity is in $\tilde{O}(\sqrt{B})$, where $\tilde{O}$ hides also polynomial factors in $\log q$.*

Proof. We use the algorithm given in [6, §3.2].

Let $b = \lfloor \sqrt{B} \rfloor$. Define two sets $I = \{1, \dots, b\}$ and $J = \{0, b, \dots, b(b-1)\}$. Then, the resultant of two polynomials $\prod_{i \in I} (X - (-p)^i)$ and $\prod_{j \in J} ((-p)^j X - 1)$ in $\mathbb{F}_q[X]$ is equal to $\prod_{k=1}^{b^2} (1 - (-p)^k) \in \mathbb{F}_q$. This resultant can be computed in $\tilde{O}(b)$ operations in $\mathbb{F}_q$. Since q divides $\prod_{k=1}^B ((-p)^k - 1)$ if and only if the resultant is equal to 0 or $\prod_{k=b^2+1}^B ((-p)^k - 1) \equiv 0 \pmod{q}$, we can check whether q divides $\prod_{k=1}^B ((-p)^k - 1)$ with expected time complexity in $\tilde{O}(\sqrt{B})$. $\square$

We denote the algorithm in Lemma 4 by $\text{IsGoodPrime}(p, q, B)$.

4.2 Heuristic Assumptions

For positive integers a and B such that $B < 2^a$, we define a subset of Primes_a as

$$\text{GoodPrimes}_{a,p,B} = \{q \in \text{Primes}_a \mid \text{IsGoodPrime}(p, q, B) = \text{True}\}.$$

We assume the following two heuristics related to this set.

Heuristic 1 *Let B be a positive integer less than 2^a . Then we assume that*

$$\#\text{GoodPrimes}_{a,p,B} \approx \frac{B}{2a \log 2}.$$

Explanation. As we explained in Section 3.3, $\#\text{Primes}_a \approx \frac{2^a}{2a \log 2}$. Let $\Phi_k(X)$ be the k -th cyclotomic polynomial. We assume that $\Phi_1(-p), \dots, \Phi_B(-p)$ are independently and uniformly distributed over $\mathbb{Z}/q\mathbb{Z}$ for all $q \in \text{Primes}_a$. For each k and q , the probability that q divides $\Phi_k(-p)$ is approximately $1/2^a$. Therefore, the expected number of primes $q \in \text{Primes}_a$ such that q divides $\Phi_k(-p)$ for some $k \in [1, B]$ is approximately

$$\frac{B}{2^a} \cdot \#\text{Primes}_a \approx \frac{B}{2a \log 2}.$$

Heuristic 2 *Let u be a positive real number in $[1, a]$ and B be a positive integer less than 2^a . Then we assume that for a prime q sampled uniformly at random from $\text{GoodPrimes}_{a,p,B}$, the probability that $2^a - q$ is $2^{a/u}$ -smooth depends only on u .*

Explanation. Let x be a positive real number. From the Dickman theorem [12, Theorem 1.4.9], the number of $x^{1/u}$ -smooth positive integers less than x is asymptotically $x\rho(u)$, where $\rho(u)$ is the Dickman function. Assuming that 2^a is sufficiently large and that $2^a - q$ behaves as a uniform random integer in $[1, 2^a)$, we can expect that the probability that $2^a - q$ is $2^{a/u}$ -smooth is approximately $\rho(u)$.

We conducted experiments to verify these heuristics. The results support our assumptions. The details of the experiments are given in Appendix B.

5 Attacks on PRISM-id

In this section, we give two classical forgery attacks on PRISM-id [3] whose expected time complexities are exponential in the security parameter, but slightly smaller than the claimed security level.

In the rest of the paper, we use the same notation as in Section 3. Let λ be the security parameter of PRISM-id. Hence the claimed expected time complexity of PRISM-id is at least proportional to 2^λ , up to polynomial factors in λ . Since PRISM-id uses $a \approx \lambda + \log \lambda$, our purpose is to show that the expected time complexity of our attacks is $\tilde{O}(2^{a\varepsilon})$ for some $\varepsilon < 1$.

We define a *forgery attack on PRISM-id* as a direct attack (Attack Game 18.1 in [8]). In this attack, the challenger and the attacker interact as follows.

1. The challenger generates a key pair $(E_{\text{vk}}, I_{\text{sk}})$ by the key generation algorithm of PRISM-id and sends E_{vk} to the attacker.
2. The challenger sends a challenge prime $q \in \text{Primes}_a$ to the attacker.
3. The attacker outputs a response φ_{rsp} .

The attacker wins if $\text{VerIsogeny}(E_{\text{vk}}, \varphi_{\text{rsp}}, q) = \text{True}$.

We also define a *forgery attack on PRISM-id with refusing challenge primes* as a forgery attack on PRISM-id in which the attacker can refuse a challenge prime q in step 2 and ask the challenger to send another challenge prime $q' \in \text{Primes}_a$.

Algorithm 1: Attack on PRISM-id

Input: A supersingular elliptic curve E , a prime q in Primes_a , and a positive real number $\varepsilon < 1/3$.
Output: An isogeny from E of degree $q(2^a - q)$.

```

1  $B_1 \leftarrow \lceil 2^{a\varepsilon} \rceil$ ;
2  $B_2 \leftarrow \max\{2^{a\varepsilon}, 2^{a(\varepsilon/3+1/6)}\}$ ;
3 if  $\text{IsGoodPrime}(p, q, B_1) = \text{False}$  or  $2^a - q$  is not  $B_2$ -smooth then
4   return Failure;
5  $k \leftarrow 1$ ;
6 while  $(-p)^k \not\equiv 1 \pmod{q}$  do
7    $k \leftarrow k + 1$ ;
8  $\mathbb{F}_{p^{2k}} \leftarrow \text{ConstructField}(p, k)$ ;
9  $\varphi \leftarrow \text{Isogeny}(E, q, \mathbb{F}_{p^{2k}})$ ;
10  $F \leftarrow$  the codomain of  $\varphi$ ;
11 Let  $\ell_1^{e_1} \cdots \ell_n^{e_n}$  be the prime factorization of  $2^a - q$ ;
12 for  $i \leftarrow 1$  to  $n$  do
13    $\mathbb{F}_{p^{2(\ell_i-1)}} \leftarrow \text{ConstructField}(p, \ell_i - 1)$ ;
14   for  $j \leftarrow 1$  to  $e_i$  do
15      $\psi \leftarrow \text{Isogeny}(F, \ell_i, \mathbb{F}_{p^{2(\ell_i-1)}})$ ;
16      $F \leftarrow$  the codomain of  $\psi$ ;
17      $\varphi \leftarrow \psi \circ \varphi$ ;
18 return  $\varphi$ ;
```

5.1 Forgery attack on PRISM-id without refusing challenge primes

We first consider an attack in which the attacker cannot refuse a challenge prime q . This attack only succeeds when the challenge prime q is a good prime. Such a good prime appears with exponentially small probability, but the probability is greater than $2^{-a} \approx 2^{-(\lambda + \log \lambda)}$.

This attack is described in Algorithm 1. First, the attacker checks whether the challenge prime q is a good prime by using $\text{IsGoodPrime}(p, q, B_1)$ for some bound B_1 and $2^a - q$ is B_2 -smooth for some bound B_2 . If q does not satisfy these conditions, the attacker fails. Otherwise, the attacker finds the smallest k such that $E[q] \subset E(\mathbb{F}_{p^{2k}})$. Then, the attacker computes a q -isogeny and ℓ -isogenies for all prime factors ℓ of $2^a - q$. Finally, the attacker outputs the composition of these isogenies as a response. The success probability and expected time complexity of this attack are given in the following theorem.

Theorem 1. *Assume Heuristic 1 and Heuristic 2. For any positive real number $\varepsilon < 1/3$, there exists a positive constant C_ε depending only on ε and a forgery attack on PRISM-id with success probability $C_\varepsilon \cdot 2^{-a(1-\varepsilon)}$ and expected time complexity $\tilde{O}(\max\{2^{3a\varepsilon}, 2^{a(\varepsilon+1/2)}\})$.*

Proof. We show the forgery attack on PRISM-id in Algorithm 1. From Heuristic 1 and Heuristic 2, we can expect that the number of primes $q \in \text{Primes}_a$ which

pass the check in line 3 is approximately $C_\varepsilon \cdot 2^{a\varepsilon} / (2a \log 2)$, where C_ε is a constant depending only on ε . Therefore, the success probability of this algorithm is approximately

$$\frac{C_\varepsilon \cdot 2^{a\varepsilon}}{2a \log 2} \cdot \frac{1}{\#\text{Primes}_a} = C_\varepsilon \cdot 2^{-a(1-\varepsilon)}.$$

The expected time complexity is the sum of those of the following procedures:

1. **ConstructField**(p, k),
2. **IsGoodPrime**(p, q, B_1),
3. the factorization of $2^a - q$,
4. finding k such that $(-p)^k \equiv 1 \pmod{q}$ (**while** loop in line 6),
5. **Isogeny**($E, q, \mathbb{F}_{p^{2k}}$),
6. **ConstructField**($p, \ell_i - 1$) for $i = 1, \dots, n$,
7. **Isogeny**($F, \ell_i, \ell_i - 1$) (e_i times) for $i = 1, \dots, n$.

As in Algorithm 1, we let $B_2 = \max\{2^{a\varepsilon}, 2^{a(\varepsilon/3+1/6)}\}$. It suffices to show that each step has the expected time complexity in $\tilde{O}(B_2^3)$.

The expected time complexity of step 1 is $\tilde{O}(2^{3a\varepsilon})$ as explained in Section 4.1. The expected time complexity of step 2 is $\tilde{O}(2^{a\varepsilon/2})$ by Lemma 4. In the factorization in step 3, we can find all prime factors of $2^a - q$ less than B_2 with expected time complexity in $\tilde{O}(\sqrt{B_2})$ by the Pollard rho algorithm [12, Algorithm 5.2.1].³ The expected time complexity of step 4 is $\tilde{O}(B_1) = \tilde{O}(2^{a\varepsilon})$ since $k \leq B_1$ and the exponentiation $(-p)^k \pmod{q}$ can be computed in $O(\log k)$ multiplications in $\mathbb{F}_q$. The expected time complexity of step 5 is $\tilde{O}(2^{a(\varepsilon+\frac{1}{2})})$ by Lemma 3. For each prime ℓ_i in the factorization of $2^a - q$, the expected time complexity of step 6 is $\tilde{O}(B_2^3)$ and the expected time complexity of step 7 is $\tilde{O}(B_2^2)$. The number $\sum_{i=1}^n e_i$ of iterations of the two **for** loops in Algorithm 1 is at most a . Therefore, the expected time complexity of Algorithm 1 is $\tilde{O}(B_2^3) = \tilde{O}(\max\{2^{3a\varepsilon}, 2^{a(\varepsilon+1/2)}\})$. $\square$

In particular, we have the following forgery attack on PRISM-id.

Corollary 1. *Assuming Heuristic 1 and Heuristic 2, there exists a forgery attack on PRISM-id with success probability $\tilde{\Theta}(2^{-3(\lambda+\log \lambda)/4})$ and expected time complexity $\tilde{O}(2^{3\lambda/4})$.*

Proof. Since $a \approx \lambda + \log \lambda$, it follows from Theorem 1 by taking $\varepsilon = 1/4$. $\square$

Remark 2. The coefficient C_ε in the success probability in Theorem 1 is approximately $\rho(a/\log_2(B_2)) = \rho((\max\{\varepsilon, \varepsilon/3+1/6\})^{-1})$, where ρ is the Dickman function. This coefficient is not so small when ε is close to $1/3$. For example, when $\varepsilon = 1/4$, we have $C_\varepsilon \approx 0.0049$.

³ ECM (elliptic curve method) [12, Algorithm 7.4.2] has subexponential expected time complexity in $\log B_2$, and is faster than the Pollard rho algorithm in most cases. However, it is enough to use the Pollard rho algorithm to prove the expected time complexity of our attack.

Algorithm 2: IsogenyFromOracle($q, t, \text{ProvingOracle}_E$)

Input: A prime q in Primes_a , a positive integer t , and a proving oracle ProvingOracle_E .

Output: An isogeny from E of degree t .

```

1  $s \leftarrow 1$ ;
2 while  $st < 2^{a-1}$  and  $2^a - st$  is not a prime in  $\text{Primes}_a \setminus \{q\}$  do
3    $s \leftarrow s + 2$ ;
4 if  $st \geq 2^{a-1}$  then
5   return Failure;
6  $\varphi \leftarrow \text{ProvingOracle}_E(2^a - st)$ ;
7  $\psi \leftarrow$  the  $st$ -isogeny from  $E$  obtained from  $\varphi$ ; // Using Kani's lemma.
8  $k \leftarrow$  the smallest positive integer such that  $(-p)^k \equiv 1 \pmod{s}$ ; //  $k < s$ .
9  $\mathbb{F}_{p^{2k}} \leftarrow \text{ConstructField}(p, k)$ ;
10  $(P, Q) \leftarrow$  a basis of  $E[s]$ ; //  $P$  and  $Q$  are in  $E(\mathbb{F}_{p^{2k}})$ .
11  $\psi_s \leftarrow$  an isogeny of degree  $s$  with kernel  $\langle \psi(P), \psi(Q) \rangle$ ; //  $\psi = \hat{\psi}_s \circ \psi_t$  for a
     $t$ -isogeny  $\psi_t$ .
12  $(P_a, Q_a) \leftarrow$  a basis of  $E[2^a]$ ;
13  $\psi_t(P_a) \leftarrow [1/s]\psi_s \circ \psi(P_a)$ ,  $\psi_t(Q_a) \leftarrow [1/s]\psi_s \circ \psi(Q_a)$ ;
14 return  $\psi_t$ ; // expressed by  $(P_a, Q_a, \psi_t(P_a), \psi_t(Q_a))$ .

```

Using a Proving Oracle. We consider an active attack (Attack Game 18.3 in [8]), where the attacker can interact with a proving oracle $\text{ProvingOracle}_{E_{\text{vk}}}(q)$ that takes a prime $q \in \text{Primes}_a$ as input and returns an efficient representation of an isogeny of degree $q(2^a - q)$ from E_{vk} .

By using a proving oracle, we can mitigate the second condition that $2^a - q$ is B_2 -smooth in the attack in Algorithm 1. Instead, we only require that $2^a - q$ is the product of a B_2 -smooth number and an integer t such that there exists a small integer s such that $2^a - st$ is a prime in Primes_a different from q . This mitigation does not improve the asymptotic success probability of the attack since the probability that $2^a - q$ is B_2 -smooth is a constant depending only on ε by Heuristic 2. However, it can improve the constant factor in the success probability. In the following, we explain the reason why we can mitigate the condition by using a proving oracle.

Let d be the largest B_2 -smooth number dividing $2^a - q$, and let $t = (2^a - q)/d$. Suppose that there exists a small integer s such that $2^a - st$ is a prime in Primes_a different from q . We can expect that such an s exists when $2^{a-1}/t > 2a \log 2$ since the probability that $2^a - st$ is a prime is approximately $1/(2a \log 2)$ for a random $s \in [1, 2^a/t]$. Then the attacker can obtain an isogeny of degree t by calling a proving oracle with the input $2^a - st$. This is done as follows: First, the attacker obtains an isogeny of degree $(2^a - st)st$ from E by calling the oracle with the input $2^a - st$. Then, the attacker computes an isogeny φ of degree st from E by using Kani's lemma. By evaluating a basis of $E[s]$ under φ , the attacker obtains generators of the kernel of degree s part of $\hat{\varphi}$. This allows the attacker to compute an isogeny of degree t from E . We describe this algorithm

Algorithm 3: Attack on PRISM-id with a proving oracle

Input: A supersingular elliptic curve E , a prime q in Primes_a , a positive real number $\varepsilon < 1/3$, and a proving oracle ProvingOracle_E .

Output: An isogeny from E of degree $q(2^a - q)$.

```

1   $B_1 \leftarrow \lceil 2^{a\varepsilon/3} \rceil$ ;
2   $B_2 \leftarrow \max\{2^{a\varepsilon}, 2^{a(\varepsilon/3+1/6)}\}$ ;
3  if  $\text{IsGoodPrime}(p, q, B_1) = \text{False}$  then
4      return Failure;
5  Let  $s$  be the largest  $B_2$ -smooth number dividing  $2^a - q$ ;
6   $t \leftarrow (2^a - q)/s$ ;
7  if  $t = 1$  then
8      Do the same as after line 5 in Algorithm 1;
9  else
10      $\varphi \leftarrow \text{IsogenyFromOracle}(q, t, \text{ProvingOracle}_E)$ ;
11     if  $\varphi = \text{Failure}$  then
12         return Failure;
13      $F \leftarrow$  the codomain of  $\varphi$ ;
14      $k \leftarrow 1$ ;
15     while  $(-p)^k \not\equiv 1 \pmod{q}$  do
16          $k \leftarrow k + 1$ ;
17      $\mathbb{F}_{p^{2k}} \leftarrow \text{ConstructField}(p, k)$ ;
18      $\psi \leftarrow \text{Isogeny}(F, q, \mathbb{F}_{p^{2k}})$ ;
19      $\varphi \leftarrow \psi \circ \varphi$ ;
20      $F \leftarrow$  the codomain of  $\varphi$ ;
21     Let  $\ell_1^{e_1} \dots \ell_n^{e_n}$  be the prime factorization of  $s$ ;
22     for  $i \leftarrow 1$  to  $n$  do
23          $\mathbb{F}_{p^{2(\ell_i-1)}} \leftarrow \text{ConstructField}(p, \ell_i - 1)$ ;
24         for  $j \leftarrow 1$  to  $e_i$  do
25              $\psi \leftarrow \text{Isogeny}(F, \ell_i, \mathbb{F}_{p^{2(\ell_i-1)}})$ ;
26              $F \leftarrow$  the codomain of  $\psi$ ;
27              $\varphi \leftarrow \psi \circ \varphi$ ;
28     return  $\varphi$ ;
```

in Algorithm 2. In the algorithm, we can expect that s is $O(a)$ as explained above. Therefore, the expected time complexity of the algorithm is a polynomial in $\log p$.

We describe the attack on PRISM-id using a proving oracle in Algorithm 3.

5.2 Forgery attack on PRISM-id with refusing challenge primes

If we allow the attacker to refuse a challenge prime q , then we can construct a forgery attack with expected time complexity in $\tilde{O}(2^{6a/7})$. This attack is obtained by refusing challenge primes until the condition in line 3 in Algorithm 3

is satisfied. At first glance, the expected time complexity of this attack seems to be greater than $2^{a(1-\varepsilon)} \cdot 2^{3a\varepsilon} = 2^{a(1+2\varepsilon)}$. However, the latter part of Algorithm 3 is executed only when the condition in line 3 is satisfied. Therefore, the expected time complexity of this attack is the sum of the expected computational complexities of obtaining a prime q passing the check in line 3 and the remaining part of the algorithm. We explicitly state this in the following theorem.

Theorem 2. *Assume Heuristic 1 and Heuristic 2. There exists a forgery attack on PRISM-id that allows the attacker to refuse a challenge prime q with expected time complexity in $\tilde{O}(2^{6a/7})$.*

Proof. Consider Algorithm 1 with $\varepsilon = 2/7$ and modify it as follows:

- q is not given as input, but sampled uniformly at random from Primes_a .
- If the condition in line 3 is not satisfied, then the algorithm resamples q uniformly at random from Primes_a and checks the condition again.

Since $B_1 = \lceil 2^{2a/7} \rceil$ in this case, the expected number of checks of the condition in line 3 is approximately $C(2a \log 2/B_1) \cdot \#\text{Primes}_a \approx C \cdot 2^{5a/7}$, where C is a constant derived from Heuristic 2. The expected time complexity of checking the condition in line 3 is in $\tilde{O}(2^{a/7})$ by Lemma 4. Therefore, the expected time complexity of finding a prime q passing the check in line 3 is in $\tilde{O}(2^{6a/7})$.

The remaining part of the algorithm is in $\tilde{O}(2^{6a/7})$ as shown in the proof of Theorem 1. This completes the proof. $\square$

5.3 Countermeasure

For PRISM-id to be secure against the forgery attacks in the previous subsections, the protocol should use a large enough a . Since the proposed p for PRISM-id in [3] is of the form $p = f2^e - 1$, where f is a small odd integer and $e \approx 2\lambda$, we can increase a as far as $a < 2\lambda$ while keeping $E[2^a]$ rational.

We propose two sizes of a for PRISM-id.

- **Conservative size:** $a = \frac{4}{3}\lambda$. This size ensures the following: In Algorithm 1 with any ε , the success probability is less than $2^{-\lambda}$ or the expected time complexity is greater than approximately 2^λ .
- **Aggressive size:** $a = \frac{7}{6}\lambda$. This size ensures that the expected time complexity of the forgery attack in §5.2 is greater than approximately 2^λ .

Note that for the aggressive size, there exists a forgery attack whose success probability is greater than $2^{-\lambda}$ and whose expected time complexity is less than 2^λ . For example, by taking $\varepsilon = 3/14$ in Algorithm 1, the success probability is proportional to $2^{-11\lambda/12}$ and the expected time complexity is $\tilde{O}(2^{5\lambda/6})$.

5.4 Quantum setting

In the quantum setting, we can slightly improve the expected time complexities of our forgery attacks by using Grover's search algorithm [23]. However, the improvement does not affect the quantum security of PRISM-id.

In our attacks, we can apply Grover's search to irreducible polynomial search in the algorithm `ConstructField`(p, k). As a result, the expected time complexity of `ConstructField`(p, k) is improved from $\tilde{O}(k^3)$ to $\tilde{O}(k^{5/2})$. This improvement reduces the expected time complexities of the attack in Theorem 1 from $\tilde{O}(\max\{2^{3a\varepsilon}, 2^{a(\varepsilon+1/2)}\})$ to $\tilde{O}(\max\{2^{5a\varepsilon/2}, 2^{a(\varepsilon+1/2)}\})$ and that of the attack in Theorem 2 from $\tilde{O}(2^{6a/7})$ to $\tilde{O}(2^{5a/6})$.

On the other hand, there are more efficient quantum algorithms to break PRISM-id. For example, the Delfs-Galbraith algorithm [17] combined with Grover's search can find an isogeny from E_0 to E_{vk} with expected time complexity in $\tilde{O}(p^{1/4})$ (see [7] for more details on the quantum algorithms for finding isogenies). Since $p \approx 2^{2\lambda}$ and $a \approx \lambda + \log_2 \lambda$ in PRISM-id, this expected time complexity is smaller than those of our forgery attacks in the quantum setting.

6 Attack on the underlying problem

In this section, we consider an attack on the underlying problem of PRISM-id (Problem 4), which is to find an isogeny of degree $q(2^a - q)$ for a prime $q \in \text{Primes}_a$. The difference from the forgery attacks in Section 5 is that the attacker can choose the prime q as they like. This allows the attacker to take $\varepsilon = 0$ in Algorithm 1, thus there exists an attack with expected time complexity in $\tilde{O}(2^{a/2})$.

Note that an attack on the underlying problem does not immediately lead to a forgery attack on PRISM-id. Therefore, the results in this section should be regarded as a warning for constructing a cryptosystem based on the hardness of Problem 4.

To describe the attack, we define the following problem.

Problem 5. Given a supersingular elliptic curve E , compute an isogeny of degree $q(2^a - q)$ from E with a prime $q \in \text{Primes}_a$.

Obviously, Problem 4 reduces to Problem 5. Our attack on Problem 5 is described in Algorithm 4. The difference from Algorithm 1 is that the attacker can choose the prime q as they like. To find a prime q such that $E[q] \subset E(\mathbb{F}_{p^{2k}})$ for some small k , the attacker uses the cyclotomic polynomials. Once the attacker finds a prime q such that $E[q] \subset E(\mathbb{F}_{p^{2k}})$ for some small k and that $2^a - q$ is smooth, the following steps are the same as those in Algorithm 1. The expected time complexity of this attack is given in the following theorem.

Theorem 3. *Assuming Heuristic 1 and Heuristic 2, there exists an algorithm to solve Problem 5 with expected time complexity $\tilde{O}(2^{a/2})$.*

Proof. We show the algorithm in Algorithm 4.

From Heuristic 1 and Heuristic 2, we can expect that the expected iteration number of the while loop in Algorithm 4 is proportional to a . Thus, k is also proportional to a . As explained in the proof of Theorem 1, finding a factor in Primes_a of $\Phi_k(-p)$ can be done in $\tilde{O}(2^{a/2})$ and finding all prime factors less than

Algorithm 4: Attack on Problem 5

Input: A supersingular elliptic curve E .
Output: An isogeny from E of degree $q(2^a - q)$ with a prime $q \in \text{Primes}_a$.

```

1  $k \leftarrow 1$ ;
2  $q \leftarrow 0$ ;
3 while  $q = 0$  do
4    $\Phi \leftarrow$  the  $k$ -th cyclotomic polynomial;
5   if The prime factors of  $\Phi(-p)$  contain a prime in  $\text{Primes}_a$  then
6      $q \leftarrow$  a prime factor of  $\Phi(-p)$  in  $\text{Primes}_a$ ;
7     if  $2^a - q$  is not  $2^{a/6}$ -smooth then
8        $q \leftarrow 0$ ;
9        $k \leftarrow k + 1$ ;
10  else
11     $k \leftarrow k + 1$ ;
12 Let  $\ell_1^{e_1} \cdots \ell_n^{e_n}$  be the prime factorization of  $2^a - q$ ;
13  $\mathbb{F}_{p^{2k}} \leftarrow \text{ConstructField}(p, k)$ ;
14  $\varphi \leftarrow \text{Isogeny}(E, q, \mathbb{F}_{p^{2k}})$ ;
15  $F \leftarrow$  the codomain of  $\varphi$ ;
16 for  $i \leftarrow 1$  to  $n$  do
17    $\mathbb{F}_{p^{2(\ell_i-1)}} \leftarrow \text{ConstructField}(p, \ell_i - 1)$ ;
18   for  $j \leftarrow 1$  to  $e_i$  do
19      $\psi \leftarrow \text{Isogeny}(F, \ell_i, \mathbb{F}_{p^{2(\ell_i-1)}})$ ;
20      $F \leftarrow$  the codomain of  $\psi$ ;
21      $\varphi \leftarrow \psi \circ \varphi$ ;
22 return  $\varphi$ ;
```

$2^{a/6}$ of $2^a - q$ can be done in $\tilde{O}(2^{a/12})$. Therefore, the expected time complexity of the while loop is in $\tilde{O}(2^{a/2})$.

The expected time complexity of $\text{ConstructField}(p, k)$ in line 13 is bounded by some polynomial in $\log p$ since k is proportional to $a < \log p$.

The isogeny φ in the algorithm is computed by the algorithm $\text{Isogeny}(E, q, \mathbb{F}_{p^{2k}})$ in Lemma 3. The expected time complexity of this algorithm is in $\tilde{O}(2^{a/2})$.

The number $\sum_{i=1}^t e_i$ of iterations of the two **for** loops in Algorithm 4 is at most a . In each iteration, the expected time complexities of $\text{ConstructField}(p, \ell_i - 1)$ and $\text{Isogeny}(F, \ell_i, \mathbb{F}_{p^{2(\ell_i-1)}})$ are in $\tilde{O}(2^{a/2})$ and $\tilde{O}(2^{a/3})$, respectively.

Therefore, the total expected time complexity of the algorithm is in $\tilde{O}(2^{a/2})$. $\square$

As in the attack on PRISM-id, we can mitigate the condition that $2^a - q$ is $2^{a/6}$ -smooth by using SPEDIO. Unlike ProvingOracle_E , we can call SPEDIO with input a supersingular elliptic curve different from E . In other words, we have $\text{ProvingOracle}_{E'}$ for all supersingular elliptic curves E' . By using SPEDIO,

the attacker could obtain a t -isogeny from E even in the case that Algorithm 2 fails.

This is done as follows: Let t be a non-smooth factor of $2^a - q$. Suppose that t decomposes into a product $t_1 \cdots t_n$ such that there exist small integers $s_1, \dots, s_n$ satisfying $2^a - s_i t_i \in \text{Primes}_a \setminus \{q\}$ for all $i = 1, \dots, n$. Then the attacker can obtain a t -isogeny from E as follows:

1. Let E_1 be E .
2. For $i = 1, \dots, n$,
 - (a) $\varphi_i \leftarrow \text{IsogenyFromOracle}(E_i, t_i, \text{ProvingOracle}_{E_i})$
 - (b) Let E_{i+1} be the codomain of φ_i .
3. Output $\varphi_n \circ \cdots \circ \varphi_2 \circ \varphi_1$.

7 Conclusion

We proposed classical forgery attacks on PRISM-id in scenarios where refusing a challenge is allowed and where it is not. In both cases, we showed that the original parameters in PRISM-id do not achieve the claimed security level. We also proposed a countermeasure against our attacks. Additionally, we analyzed the underlying problem of PRISM-id and gave an explicit expected time complexity for solving it. We believe that our work provides valuable insights into the security of PRISM-id and is useful for constructing new protocols based on the hardness of the given-degree isogeny problem.

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A Proof of Lemma 3 for the General Case

We give a proof of Lemma 3 for the general case. In particular, we allow p and q to be arbitrary primes distinct from each other.

Let ψ be the second Chebyshev function, i.e.,

$$\psi(x) = \sum_{k \in \mathbb{N}} \sum_{\ell^k \leq x} \log \ell = \sum_{\ell \leq x} [\log_{\ell} x] \log \ell,$$

where ℓ runs over all primes. From Theorem 414 in [24], there exists a constant C such that $\psi(x) > Cx$ for all $x > 0$. This means that there exists a $(\log q)/C$ -powersmooth number greater than q . Let $\ell_1^{e_1} \dots \ell_n^{e_n}$ be the prime factorization of this number.

For $i = 1, \dots, n$, let (R_i, S_i) be a basis of $E[\ell_i^{e_i}]$. Note that $E[\ell_i^{e_i}]$ is rational over the extension field of $\mathbb{F}_{p^2}$ of degree $(\ell_i - 1)\ell_i^{e_i-1}$. Since $\ell_i^{e_i} \leq (\log q)/C$ and $n < \log_2 q$, we can compute (R_i, S_i) for all $i = 1, \dots, n$ with expected time complexity in polynomial in $\log p$ and $\log q$.

We use these bases $\{(R_i, S_i)\}_i$ instead of (P_a, Q_a) in the proof of Lemma 3 for the special case. The images $\varphi(R_i)$ and $\varphi(S_i)$ under the q -isogeny in step 4 can also be computed in $\tilde{O}(k\sqrt{q})$. Therefore, the expected time complexity of the algorithm for the general case is also in $\tilde{O}(\max\{k^2, k\sqrt{q}\})$.

As in the proof of the special case, we let E' be the codomain of φ . Then, $(E', q, \{R_i, S_i, \varphi(R_i), \varphi(S_i)\}_i)$ is an efficient representation of φ with respect to

the 8-dimensional isogeny computation in [35] since $(\sum_i R_i, \sum_i S_i)$ is a basis of $E[\prod_i \ell_i^{e_i}]$. Note that we do not compute this basis explicitly and the computation can be done in polynomial time by using a similar approach as in [21, Algorithm 2].

B Experiments to confirm heuristics

We show experimental results to confirm Heuristic 1 and Heuristic 2.

B.1 Experiments for Heuristic 1

We show experimental results to confirm Heuristic 1. The setting in our experiment is as follows: For a given a , we chose a prime p of the form $2^{2a-1}f - 1$, where f is the smallest positive integer such that p is a prime number.

Then, we conducted two experiments:

1. We computed the cardinality of $\text{GoodPrimes}_{a,p,B}$ in Heuristic 1 for $B = 1, \dots, 2^a - 1$.
2. We searched for the prime factors of $\Phi_k(-p)$ in Primes_a for $k = 1, \dots, 4a$.

The first experiment is conducted for $a = 10, 20$. The results are shown in Figure 1. In the figure, the horizontal axis is B and the vertical axis is the cardinality of $\text{GoodPrimes}_{a,p,B}$. The blue line is the exact cardinality of $\text{GoodPrimes}_{a,p,B}$, and the orange line is the number predicted by Heuristic 1. The results show that there are slightly more good primes than predicted by Heuristic 1.

In the second experiment, we chose $a = 30, 40, 50$. For searching for the prime factors of $\Phi_k(-p)$ in Primes_a , we use the Pollard rho algorithm. The results are shown in Table 2. The number of the prime factors of $\Phi_k(-p)$ in Primes_a for $k = 1, \dots, 4a$ is equal to the cardinality of $\text{GoodPrimes}_{a,p,4a}$. Therefore, Heuristic 1 predicts that the number is $2/\log 2 \approx 2.8854$. The results almost match the expected number.

We also confirm that a proving oracle can help the attacker to obtain an isogeny of degree t , which is a non-smooth factor of $2^a - q$. The results are shown in Table 3. There are four cases where the search for another $q' \in \text{Primes}_a$ fails. In these cases, the attack on PRISM-id with a proving oracle should search for a different prime q . On the other hand, the attack on Problem 4 can succeed in some of these cases by using SPEDIO as explained in Section 6. Indeed, except for the case of $q = 876847801$, we can find q' by decomposing a non-smooth factor t of $2^a - q$.

B.2 Experiments for Heuristic 2

To confirm Heuristic 2, we computed the ratio

$$\frac{\#\{q \in \text{GoodPrimes}_{a,p,B} \mid 2^a - q \text{ is } (2^{a/u})\text{-smooth}\}}{\#\text{GoodPrimes}_{a,p,B}}$$

by listing all the primes $q \in \text{GoodPrimes}_{a,p,B}$ and checking if $2^a - q$ is $(2^{a/u})$ -smooth. As explained in Section 4.2, we expect this ratio to be close to the Dickman function $\rho(u)$.

We chose $a = 30$ and $p = 2^{61} - 1$, as in the previous experiments. The bound B was set to $B_1 = \lceil 2^{a\varepsilon} \rceil$ as in Algorithm 1 for $\varepsilon \in \{\frac{1}{4}, \frac{2}{7}, \frac{1}{3}\}$. Note that the first two values of ε are used in Corollary 1 and Theorem 2, respectively, and that the last value is the upper bound in Theorem 1. Then, we computed the above ratio for u in steps 0.1 over the interval $[1, 5]$. The results are shown in Figure 2. As predicted by Heuristic 2, the ratio is close to the Dickman function $\rho(u)$.

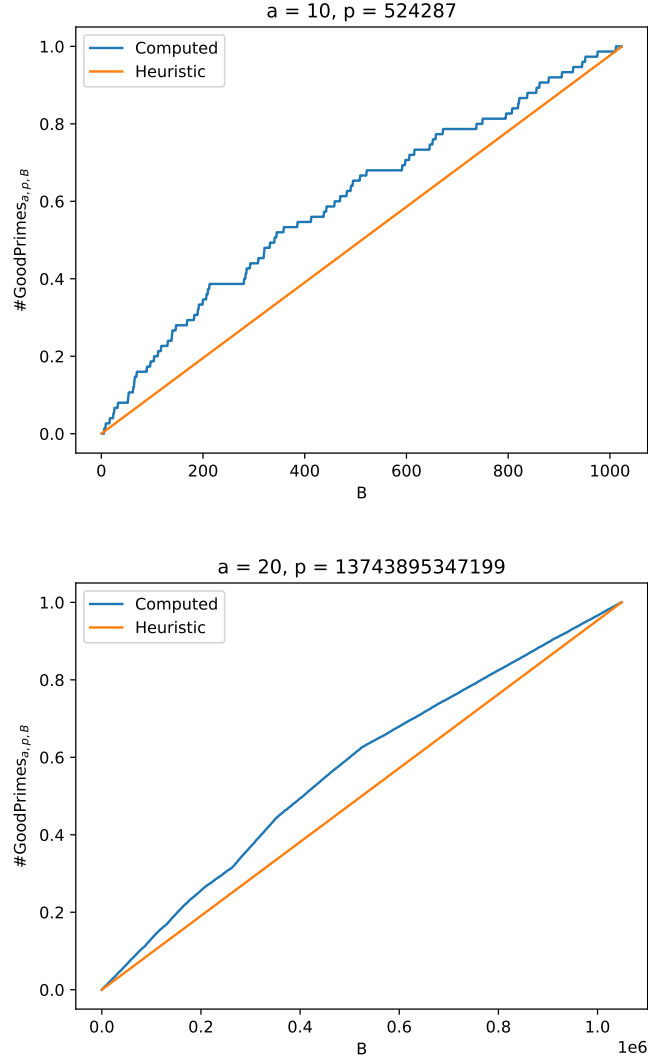


Fig. 1. The cardinality of $\text{GoodPrimes}_{a,p,B}$ for $a = 10, 20$ and $B \in [1, 2^a - 1]$. The horizontal axis is B , and the vertical axis is the cardinality of $\text{GoodPrimes}_{a,p,B}$. The blue line is the computed cardinality of $\text{GoodPrimes}_{a,p,B}$, and the orange line is the cardinality predicted by Heuristic 1.

Table 2. Integers k such that $\Phi_k(-p)$ has a prime factor q in Primes_a and the factorization of $2^a - q$. The factors in red are prime factors greater than $2^{a/6}$.

a	p	k	q s.t. $q \mid \Phi_k(-p)$	$2^a - q$
30	$2^{61} - 1$	5	625343471	$13 \cdot 31 \cdot \mathbf{1112651}$
		62	715827883	$3 \cdot 7 \cdot 11 \cdot 31 \cdot \mathbf{151} \cdot \mathbf{331}$
		100	876847801	$3 \cdot \mathbf{65631341}$
		110	627381701	$11 \cdot \mathbf{109} \cdot \mathbf{372277}$
40	$2^{80} \cdot 15 - 1$	81	851187842929	$3 \cdot \mathbf{577} \cdot \mathbf{143456837}$
		95	599173923731	$5 \cdot \mathbf{479} \cdot \mathbf{14033} \cdot \mathbf{14887}$
		160	710949960001	$3 \cdot 5^2 \cdot \mathbf{5180822237}$
50	$2^{99} \cdot 9 - 1$	75	815037005161201	$3^2 \cdot 7 \cdot 13 \cdot 263 \cdot \mathbf{1443209059}$
		170	710540400578051	$13 \cdot \mathbf{461} \cdot \mathbf{2039} \cdot \mathbf{33990899}$
		183	756997340934253	$3^2 \cdot 11 \cdot \mathbf{3726288544529}$

Table 3. Given $q \in \text{Primes}_a$ and t a non-smooth factor of $2^a - q$, the table shows the result of searching s such that $q' := 2^a - st$ is a prime in Primes_a .

a	q	t	s	q'
30	625343471	1112651	$3 \cdot 5^2$	990292999
	715827883	$151 \cdot 331$	3	1073591881
	876847801	65631341	Failure	—
	627381701	$109 \cdot 372277$	Failure	—
		109	$3 \cdot 5$	1073740189
		372277	$5 \cdot 13$	1049543819
40	851187842929	$577 \cdot 143456837$	Failure	—
		577	$3 \cdot 5$	1099511619121
		143456837	3^2	1098220516243
	599173923731	$479 \cdot 14033 \cdot 14887$	3	799309005349
	710949960001	5180822237	19	1001076005273
50	815037005161201	1443209059	3	1125895577215447
	710540400578051	$461 \cdot 2039 \cdot 33990899$	Failure	—
		$461 \cdot 2039$	$3 \cdot 19$	1125899853263821
		33990899	3	1125899804869927
	756997340934253	3726288544529	49	943311768160703

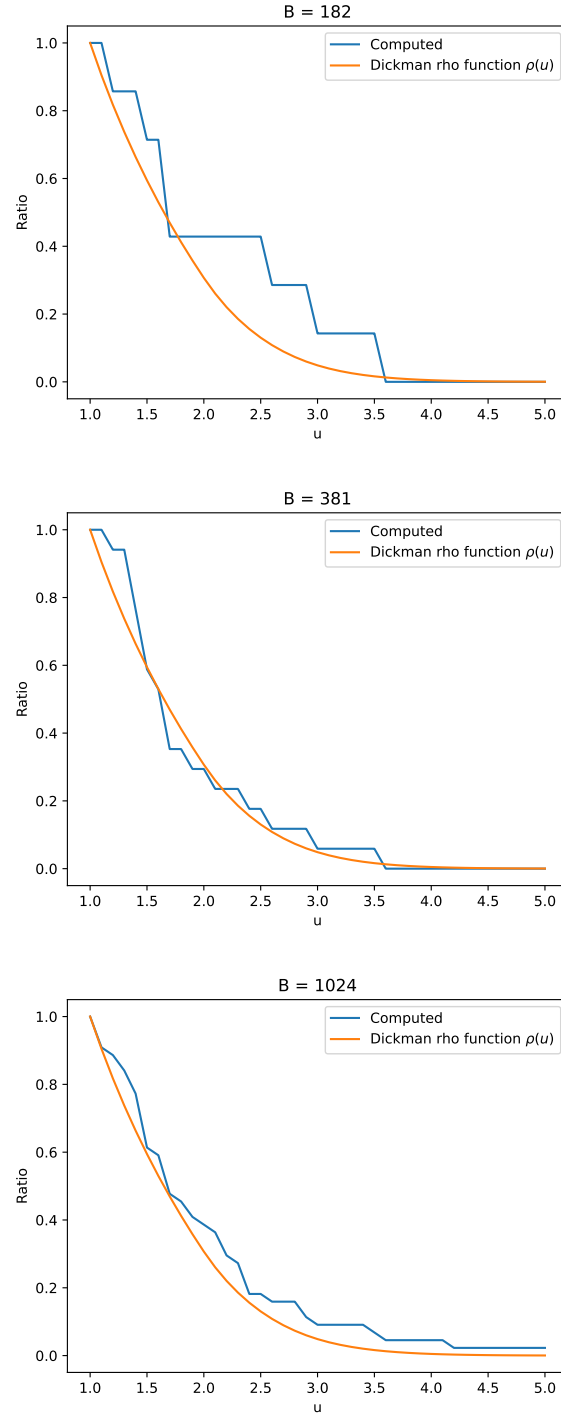


Fig. 2. The ratio of the number of $q \in \text{GoodPrimes}_{a,p,B}$ such that $2^a - q$ is $(2^{a/u})$ -smooth to the cardinality of $\text{GoodPrimes}_{a,p,B}$ for $a = 30$ and $B = B_1 = \lceil 2^{a\varepsilon} \rceil$ with $\varepsilon = 1/4$ (top), $\varepsilon = 2/7$ (middle), and $\varepsilon = 1/3$ (bottom). The horizontal axis is u , and the vertical axis is the ratio. The blue line is the computed ratio, and the orange line is the Dickman function $\rho(u)$.